

QBITEL BRIDGE

HEALTHCARE DEPLOYMENT CHECKLIST

10-Phase Implementation Guide -- Healthcare & Medical Devices



11 Phases | 200+ Checklist Items | 20-22 Week Timeline | Zero Clinical Disruption Guaranteed

Color Legend: Gold bar = Critical item | Teal bar = Standard item | Green bar = Verification item

Facility: _____ Project Lead: _____ Start Date: _____

HOW TO USE THIS CHECKLIST

Work through each phase sequentially. Obtain sign-off at the end of each phase before advancing. Gold left bar = Critical item requiring CISO or Compliance Officer sign-off. Teal left bar = Standard item (IT/Security team sign-off). Green left bar = Verification item (requires testing evidence). Owner tags: Security = Security Team, Clinical = Clinical Informatics, Biomedical = Biomedical Engineering.

0 PHASE 0: PRE-ENGAGEMENT	
Week 1-2	Owner: CISO + Project Manager
☐ 0.1.	Identify executive sponsor (CISO or VP of IT) with budget authority CISO
☐ 0.2.	Assemble core project team: CISO, CIO delegate, HIPAA Officer, Biomedical Engineering lead CISO
☐ 0.3.	Schedule kickoff meeting with QBITEL Clinical Security Engineering team PM
☐ 0.4.	Review existing medical device inventory for completeness (biomedical asset register) Biomedical
☐ 0.5.	Identify all clinical network segments and VLAN architecture documentation Network
☐ 0.6.	Collect existing HIPAA Security Rule gap assessment (if available) Compliance
☐ 0.7.	Identify all EHR and clinical application vendors (Epic, Cerner, etc.) IT
☐ 0.8.	Obtain existing network diagrams including clinical device VLANs Network
☐ 0.9.	Brief Legal team on QBITEL BAA requirements -- obtain preliminary approval Legal
☐ 0.10.	Notify FDA Cybersecurity point of contact if facility is research/regulated Compliance
☐ 0.11.	Schedule Biomedical Engineering availability for device inventory phase Biomedical
☐ 0.12.	Identify Change Control Board (CAB) process for network changes IT
Phase 0 Complete: Pre-Engagement 12 checklist items 3 critical items Sign-off required before advancing to next phase ☐ Sign-off: _____	

1

PHASE 1: MEDICAL DEVICE DISCOVERY

Week 1-2

Owner: Security Team + Biomedical Engineering

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|--------------------------|--|------------|
| ☐ | 1.1. Deploy QBITEL passive network sensors on clinical network segments
<i>Passive only -- no traffic injection, no device impact</i> | Security |
| ☐ | 1.2. Validate sensor placement covers all clinical VLANs: ICU, ED, OR, Radiology, Pharmacy, Med-Surg | Network |
| ☐ | 1.3. Run 48-hour passive discovery scan to observe all device communication patterns
<i>Do not disturb clinical operations</i> | Security |
| ☐ | 1.4. Export device inventory report and compare to biomedical asset register
<i>Expect 15-20% devices not in asset register</i> | Biomedical |
| ☐ | 1.5. Classify all discovered devices by class: imaging, monitoring, infusion, lab, other | Biomedical |
| ☐ | 1.6. Identify devices on end-of-life operating systems (Windows XP, CE, embedded Linux) | Security |
| ☐ | 1.7. Document all communication protocols observed per device class
<i>HL7, DICOM, proprietary, etc.</i> | Security |
| ☐ | 1.8. Identify devices with external internet connectivity (even occasional) | Security |
| ☐ | 1.9. Map all HL7 interface connections to EHR integration engine | IT |
| ☐ | 1.10. Map all DICOM connections to PACS and modality worklist servers | IT |
| ☐ | 1.11. Identify devices with default or shared credentials (vulnerability scan results) | Security |
| ☐ | 1.12. Generate vulnerability score report per device class for executive review
<i>Board-ready risk report delivered by QBITEL</i> | Security |
| ☐ | 1.13. Biomedical Engineering review of device inventory for accuracy | Biomedical |
| ☐ | 1.14. Document devices requiring Protocol Proxy Mode (legacy/implantable programmer) | Security |

Phase 1 Complete: Medical Device Discovery

14 checklist items | 5 critical items | Sign-off required before advancing to next phase

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Sign-off: _____

2

PHASE 2: INFRASTRUCTURE READINESS

Week 3-4

Owner: Network + Security Team

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|--------------------------|---|----------|
| ☐ | 2.1. Procure and rack QBITEL Clinical Edge Node hardware in primary data center
<i>2.1. FIPS 140-3 Level 3 HSM included in appliance</i> | IT |
| ☐ | 2.2. Procure and rack redundant (hot-standby) Clinical Edge Node | IT |
| ☐ | 2.3. Configure Bridge management VLAN (isolated from clinical VLANs) | Network |
| ☐ | 2.4. Configure 10GbE or 25GbE uplinks from Edge Nodes to core switching | Network |
| ☐ | 2.5. Configure network SPAN/tap ports on all clinical access switches
<i>2.5. Passive tap -- no traffic disruption</i> | Network |
| ☐ | 2.6. Implement VLAN segmentation per clinical area if not already in place
<i>2.6. ICU, ED, OR, Radiology, Pharmacy each in isolated VLANs</i> | Network |
| ☐ | 2.7. Configure HSM initialization and key generation on primary Edge Node
<i>2.7. Requires QBITEL HSM Administrator ceremony</i> | Security |
| ☐ | 2.8. Synchronize HSM state to hot-standby Edge Node | Security |
| ☐ | 2.9. Configure firewall rules for Bridge management traffic
<i>2.9. Port list provided by QBITEL</i> | Security |
| ☐ | 2.10. Test fail-open configuration for life-critical device VLANs (ICU, CCU, OR)
<i>2.10. CRITICAL: Life-safety devices must fail-open</i> | Security |
| ☐ | 2.11. Configure NTP synchronization for audit log timestamping | IT |
| ☐ | 2.12. Validate storage capacity for 90-day audit log retention | IT |
| ☐ | 2.13. Configure backup and disaster recovery for Edge Node configuration | IT |
| ☐ | 2.14. Submit network changes through Change Control Board (CAB) | IT |

Phase 2 Complete: Infrastructure Readiness

14 checklist items | 5 critical items | Sign-off required before advancing to next phase

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Sign-off: _____

3

PHASE 3: POLICY CONFIGURATION

Week 5-6

Owner: HIPAA Officer + Security Team

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|--------------------------|---|------------|
| ☐ | 3.1. Execute QBITEL Business Associate Agreement (BAA) -- Legal sign-off | Legal |
| ☐ | 3.2. Configure HIPAA Security Rule policy mapping in Bridge Compliance Engine
<i>45 CFR 164.312 all subsections</i> | Compliance |
| ☐ | 3.3. Define PHI data classification tiers (high, medium, standard) | Compliance |
| ☐ | 3.4. Configure PHI transmission logging policies per data classification | Security |
| ☐ | 3.5. Map device classes to clinical urgency tiers (critical, routine, background)
<i>Life-safety devices always Priority 0</i> | Clinical |
| ☐ | 3.6. Configure Battery-Aware Scheduler policies for wireless device classes | Security |
| ☐ | 3.7. Define anomaly detection thresholds per device class
<i>Use QBITEL recommended baselines initially</i> | Security |
| ☐ | 3.8. Configure VLAN quarantine automation for compromised device response
<i>Test in lab environment before production</i> | Security |
| ☐ | 3.9. Define incident notification escalation tree (Security > CISO > CEO path) | Security |
| ☐ | 3.10. Configure audit log retention policy (90-day minimum, 1-year recommended) | Compliance |
| ☐ | 3.11. Define access control policies for Bridge management console | Security |
| ☐ | 3.12. Configure role-based access for Compliance dashboard (auditor read-only) | Compliance |

Phase 3 Complete: Policy Configuration

12 checklist items | 5 critical items | Sign-off required before advancing to next phase

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Sign-off: _____

4

PHASE 4: MEDICAL DEVICE PROTECTION

Week 7-10

Owner: Security Team + Biomedical Engineering

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|--------------------------|--|------------|
| ☐ | Begin device onboarding with lowest-risk device class (lab analyzers recommended)
4.1. <i>Do not start with life-critical devices</i> | Security |
| ☐ | Deploy non-invasive wrapper on laboratory device VLAN -- validate no disruption
4.2. <i>48-hour monitoring period before expanding</i> | Security |
| ☐ | 4.3. Deploy wrapper on pharmacy automation devices | Security |
| ☐ | 4.4. Deploy wrapper on administrative medical devices (nurse call, wandering) | Security |
| ☐ | 4.5. Deploy wrapper on radiology ancillary devices (workstations, printers) | Security |
| ☐ | Deploy wrapper on patient monitoring devices (Med-Surg) -- notify unit managers
4.6. <i>Notify nursing leadership before activation</i> | Security |
| ☐ | Deploy wrapper on infusion pump fleet -- validate alarm transmission unaffected
4.7. <i>Test critical alarm delivery in monitored environment</i> | Biomedical |
| ☐ | Deploy wrapper on ICU monitoring devices -- Biomedical Engineering on-site
4.8. <i>ICU deployment requires Biomedical presence</i> | Biomedical |
| ☐ | 4.9. Deploy wrapper on ED monitoring devices | Biomedical |
| ☐ | Deploy wrapper on OR-connected devices -- coordinate with OR leadership
4.10. <i>OR blackout periods: no changes during surgical lists</i> | Clinical |
| ☐ | 4.11. Deploy wrapper on imaging modalities (CT, MRI, X-ray, Ultrasound) | Biomedical |
| ☐ | 4.12. Configure Protocol Proxy Mode for implantable device programmers | Security |
| ☐ | 4.13. Validate 100% device coverage in device inventory dashboard | Security |
| ☐ | 4.14. Biomedical Engineering sign-off on all device class deployments | Biomedical |
| ☐ | 4.15. Run 72-hour stability monitoring across all protected device VLANs | Security |

Phase 4 Complete: Medical Device Protection

15 checklist items | 7 critical items | Sign-off required before advancing to next phase



Sign-off: _____

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PHASE 5: HL7/FHIR/DICOM SECURITY

Week 9-12

Owner: IT + Integration Team

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|--------------------------|---|-----------|
| ☐ | 5.1. Inventory all HL7 v2 interface connections from EHR integration engine | IT |
| ☐ | 5.2. Deploy Bridge HL7 Protocol Security Layer on integration engine outputs
<i>Mirth Connect, Rhapsody, or Ensemble</i> | IT |
| ☐ | 5.3. Validate HL7 ADT message delivery to downstream systems (post-Bridge)
<i>Test ADT A01-A40 message types</i> | IT |
| ☐ | 5.4. Validate HL7 ORM/ORU lab order and result message delivery | IT |
| ☐ | 5.5. Deploy Bridge FHIR R4 security layer on all FHIR API endpoints | IT |
| ☐ | 5.6. Validate SMART on FHIR token issuance and validation (post-Bridge) | IT |
| ☐ | 5.7. Configure DICOM TLS overlay on all imaging modality connections | IT |
| ☐ | 5.8. Validate DICOM C-STORE operations to PACS (post-Bridge)
<i>Validate with one study from each modality type</i> | Radiology |
| ☐ | 5.9. Validate DICOM C-FIND and C-MOVE retrieval operations | Radiology |
| ☐ | 5.10. Validate Modality Worklist (MWL) delivery to imaging modalities | Radiology |
| ☐ | 5.11. Deploy X12 EDI security layer on clearinghouse connections (if applicable)
<i>837/835 transaction flows</i> | IT |
| ☐ | 5.12. Configure ML-DSA digital signatures on all outbound HL7 clinical messages
<i>Non-repudiation for clinical messages</i> | Security |
| ☐ | 5.13. Validate DICOM web services (WADO-RS, STOW-RS) -- post-Bridge | IT |
| ☐ | 5.14. Run 48-hour protocol monitoring to confirm zero message delivery failures | IT |

Phase 5 Complete: HL7/FHIR/DICOM Security

14 checklist items | 2 critical items | Sign-off required before advancing to next phase

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Sign-off: _____

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PHASE 6: EHR/HIS INTEGRATION

Week 11-14

Owner: IT + Clinical Informatics

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|--------------------------|---|------------|
| ☐ | 6.1. Configure Bridge connector for Epic Interconnect API security | IT |
| ☐ | 6.2. Validate Epic MyChart FHIR API calls through Bridge (post-activation) | IT |
| ☐ | 6.3. Configure Epic Cosmos data pipeline security (if applicable) | IT |
| ☐ | 6.4. Configure Bridge connector for Cerner Millennium API (if applicable) | IT |
| ☐ | 6.5. Configure Bridge connector for MEDITECH Expanse (if applicable) | IT |
| ☐ | 6.6. Enable PHI access audit logging in Bridge for all EHR API calls | Compliance |
| ☐ | 6.7. Configure patient consent tracking for FHIR data sharing (if applicable)
<i>Required for Da Vinci implementation guides</i> | Compliance |
| ☐ | 6.8. Validate clinical analytics platform data feeds through Bridge
<i>Health Catalyst, Arcadia, or equivalent</i> | IT |
| ☐ | 6.9. Configure HIPAA audit trail for all PHI access events | Compliance |
| ☐ | 6.10. Validate audit trail completeness with Compliance Officer | Compliance |
| ☐ | 6.11. Generate first HIPAA compliance report from Bridge Compliance Engine
<i>Validate report covers all 45 CFR 164.312 requirements</i> | Compliance |
| ☐ | 6.12. Review HIPAA report with HIPAA Security Officer -- obtain sign-off | Compliance |

Phase 6 Complete: EHR/HIS Integration

12 checklist items | 4 critical items | Sign-off required before advancing to next phase

☐ Sign-off: _____

7

PHASE 7: MONITORING & ALERTING

Week 13-16

Owner: Security Team + SIEM Team

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|--------------------------|--|------------|
| ☐ | 7.1. Activate clinical anomaly detection on all protected device VLANs | Security |
| ☐ | 7.2. Establish device communication baselines (allow 7-day learning period)
<i>Do not tune thresholds during learning period</i> | Security |
| ☐ | 7.3. Configure PHI access volume anomaly alerts | Security |
| ☐ | 7.4. Configure device-to-new-destination alerts | Security |
| ☐ | 7.5. Configure lateral movement detection (cross-VLAN anomalies) | Security |
| ☐ | 7.6. Configure VLAN quarantine automation and test in lab (non-production)
<i>Validate automated quarantine does not affect other devices</i> | Security |
| ☐ | 7.7. Integrate Bridge alerts with SIEM (Splunk, Sentinel, QRadar) | Security |
| ☐ | 7.8. Configure HIPAA breach detection alerts (PHI exfiltration indicators) | Compliance |
| ☐ | 7.9. Configure ransomware precursor detection alerts | Security |
| ☐ | 7.10. Test alert delivery to on-call security team (end-to-end test) | Security |
| ☐ | 7.11. Configure Biomedical Engineering work order integration for quarantined devices | Biomedical |
| ☐ | 7.12. Configure executive security digest reports (weekly/monthly) | Security |
| ☐ | 7.13. Run tabletop exercise simulating compromised infusion pump detection | Security |
| ☐ | 7.14. Document SOC runbook for medical device security incidents | Security |

Phase 7 Complete: Monitoring & Alerting

14 checklist items | 4 critical items | Sign-off required before advancing to next phase

☐ Sign-off: _____

8

PHASE 8: COMPLIANCE VALIDATION

Week 15-18

Owner: Compliance Officer + Security Team

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|--------------------------|---|------------|
| ☐ | 8.1. Generate full HIPAA 45 CFR 164.312 audit report -- validate all controls covered | Compliance |
| ☐ | 8.2. Review HIPAA audit report with external HIPAA counsel | Legal |
| ☐ | 8.3. Submit HIPAA audit report to Compliance Committee for approval | Compliance |
| ☐ | 8.4. Generate FDA cybersecurity documentation (SBOM) for all protected device classes | Compliance |
| ☐ | 8.5. Review FDA documentation with Biomedical Engineering and Risk Management | Biomedical |
| ☐ | 8.6. Generate HITRUST CSF evidence package for all applicable controls
<i>If HITRUST certification in scope</i> | Compliance |
| ☐ | 8.7. Review HITRUST evidence with HITRUST assessor (if applicable) | Compliance |
| ☐ | 8.8. Generate SOC 2 Type II evidence for Security and Availability trust services | Compliance |
| ☐ | 8.9. Validate breach detection capability: simulate PHI exfiltration event
<i>Confirm 15-minute detection SLA</i> | Security |
| ☐ | 8.10. Test HIPAA breach notification report generation (less than 10 minutes) | Compliance |
| ☐ | 8.11. Validate 21 CFR Part 11 electronic signature compliance for audit records | Compliance |
| ☐ | 8.12. Document post-quantum cryptography algorithm implementation for compliance records
<i>FIPS 203/204/205 documentation</i> | Security |
| ☐ | 8.13. Obtain Compliance Officer sign-off on all compliance documentation | Compliance |

Phase 8 Complete: Compliance Validation

13 checklist items | 6 critical items | Sign-off required before advancing to next phase

☐ Sign-off: _____

9

PHASE 9: CLINICAL USER ACCEPTANCE TESTING

Week 17-20

Owner: Clinical Informatics + Biomedical Engineering

☐	9.1. Brief clinical leadership (CNO, CMO, department heads) on Bridge deployment	Clinical
☐	9.2. Define clinical test scenarios per department (ICU, ED, OR, Radiology)	Clinical
☐	9.3. Conduct ICU clinical workflow validation -- nursing and intensivist sign-off <i>All alarm delivery tested</i>	Clinical
☐	9.4. Conduct ED clinical workflow validation	Clinical
☐	9.5. Conduct OR clinical workflow validation -- Anesthesia and OR nursing sign-off <i>Coordinate with OR schedule; no active cases</i>	Clinical
☐	9.6. Conduct Radiology workflow validation -- validate PACS performance unchanged <i>Timing comparison: pre- and post-Bridge image load times</i>	Radiology
☐	9.7. Conduct Pharmacy automation workflow validation	Clinical
☐	9.8. Conduct Laboratory workflow validation	Clinical
☐	9.9. Validate infusion pump alarm delivery in clinical test environment <i>Test all alarm types: occlusion, air, low battery, dose limit</i>	Biomedical
☐	9.10. Validate ventilator alarm delivery	Biomedical
☐	9.11. Validate cardiac monitor alarm delivery (ICU and telemetry)	Biomedical
☐	9.12. Document any clinical workflow concerns and resolve before go-live	Clinical
☐	9.13. Obtain written sign-off from Biomedical Engineering Director	Biomedical
☐	9.14. Obtain written sign-off from Chief Nursing Officer or delegate	Clinical
☐	9.15. Obtain written sign-off from Chief Medical Officer or delegate	Clinical

Phase 9 Complete: Clinical UAT

15 checklist items | 9 critical items | Sign-off required before advancing to next phase



Sign-off: _____

10

PHASE 10: GO-LIVE & HANDOVER

Week 19-22

Owner: All Stakeholders

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|--------------------------|--|------------|
| ☐ | 10.1. Confirm all Phase 0-9 sign-offs complete before scheduling go-live | PM |
| ☐ | 10.2. Schedule go-live during low-census period (weekend, holiday week) | PM |
| ☐ | 10.3. Notify all department heads and clinical staff of go-live date/time | PM |
| ☐ | 10.4. Confirm QBITEL Clinical Security Engineering team on-site for go-live | PM |
| ☐ | 10.5. Confirm Biomedical Engineering on-site during go-live window | Biomedical |
| ☐ | 10.6. Activate production Bridge configuration on primary Edge Node | Security |
| ☐ | 10.7. Monitor all clinical device VLANs for first 4 hours post-activation | Security |
| ☐ | 10.8. Validate anomaly detection is active and generating baseline alerts | Security |
| ☐ | 10.9. Confirm SIEM integration is receiving Bridge events | Security |
| ☐ | 10.10. Conduct 24-hour post-go-live review call with clinical leadership | PM |
| ☐ | 10.11. Handover Bridge management to internal security team (with training) | Security |
| ☐ | 10.12. Confirm 24/7 CSOC monitoring is active (QBITEL CSOC) | Security |
| ☐ | 10.13. Schedule 30-day post-deployment review meeting | PM |
| ☐ | 10.14. Generate first production HIPAA compliance report for Compliance Officer | Compliance |
| ☐ | 10.15. Submit Bridge deployment documentation to CISO and Board security committee | Security |
| ☐ | 10.16. Notify cyber insurance carrier of quantum-safe deployment for premium review
<i>Premium reduction typically 20-35%</i> | Legal |

Phase 10 Complete: Go-Live & Handover

16 checklist items | 7 critical items | Sign-off required before advancing to next phase

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Sign-off: _____

DEPLOYMENT COMPLETE

Congratulations. Your organization has achieved quantum-safe PHI protection for all connected medical devices, clinical protocols, and EHR integrations.

Phase	Name	Items	Critical	Sign-off
0	Pre-Engagement	12	3	_____
1	Medical Device Discovery	14	5	_____
2	Infrastructure Readiness	14	5	_____
3	Policy Configuration	12	5	_____
4	Medical Device Protection	15	7	_____
5	HL7/FHIR/DICOM Security	14	2	_____
6	EHR/HIS Integration	12	4	_____
7	Monitoring & Alerting	14	4	_____
8	Compliance Validation	13	6	_____
9	Clinical UAT	15	9	_____
10	Go-Live & Handover	16	7	_____
TOTAL		151+	57	_____

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